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From QFX5100 to QFX5120

An HPE Juniper Networking Techpost

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1 From QFX5100 to QFX5120

Nicole Henry - 08/27/2024

Explore the software and hardware differences you will encounter regarding switch connectivity when transitioning from the end-of-life QFX5100-48S and QFX5100-48T switches to the replacement QFX5120-48T and QFX5120-48Y switches.

This document provides an overview of the connectivity differences between the -48S and -48T QFX5100 switches and the -48Y and -48T QFX5120 switches. It also covers physical interface hardware, interface configuration, optics, and design considerations that differ between these switches.

1.1 Introduction

In April of 2021, the End Of Life (EOL) for the QFX5100-48S and QFX5100-48T switches was announced, with the last order of these switches having been on the first of July, 2022. The QFX5100 line of switches were popular switches in the leaf and access roles, and have been replaced by the QFX5120 line of switches.

The QFX5120-48Y switch replaces the QFX5100-48S, while the QFX5120-48T replaces the QFX5100-48T.

While the QFX5120 line of switches is positioned as a direct replacement for the QFX5100 line of switches, there are important differences between them, especially for the -48(x) switches in these lines. This document will not cover all differences between these switches but should provide the information you need to plan a replacement of your QFX5100-48S and -48T switches with their 5120 successors.

Readers are encouraged to review the information on the QFX5100 switches and the QFX5120 switches in the Juniper Pathfinder app.

1.2 QFX5100 to QFX5120: Hardware

1.2.1 From QFX5100-48S to QFX5120-48Y

The QFX5100-48S has 48 SFP+ downlink ports which are capable of 10 Gbps and 1 Gbps speeds using SFP+ modules for 10 GbE and SFP modules for 1 GbE. The QFX4100-48s also has 6 QSFP+ uplink ports capable of 40 Gbps speeds using QSFP+ modules.

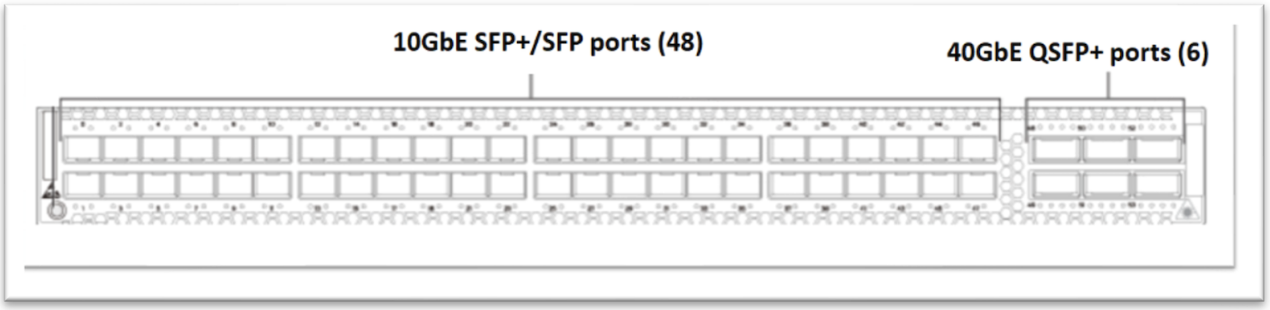


Figure 1: QFX5100-48S

The QFX5120-48Y has 48 SFP28 downlink ports that support 25 Gbps, 10 Gbps, and 1 Gbps speeds using SFP28 modules for 25 GbE, SFP+ modules for 10 Gbps, and SFP modules for 1 Gbps speeds. The QFX5120-48Y

also has 8 uplink QSFP28 ports that support 100 Gbps & 40 Gbps speeds using QSFP28 modules for 100 GbE and QSFP+ modules for 40 Gbps speeds.

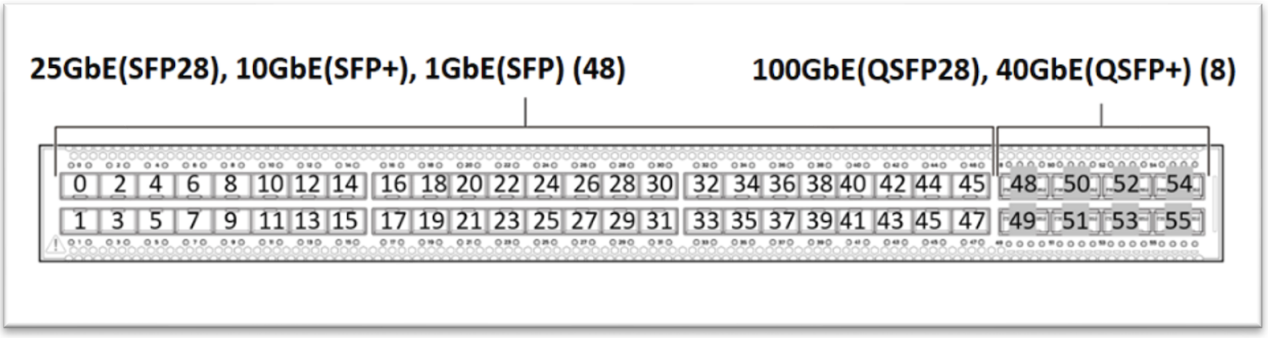


Figure 2: QFX5120-48Y

On the QFX5120-48Y, the default speed of the SFP28 interfaces (ports 0-47) is 10 Gbps, and the default speed of the QSFP28 interfaces (ports 48-55) is 100 Gbps. Additionally, the QSFP28 interfaces (ports 48-55) can be channelized to 4x25 Gbps and 4x10 Gbps.

> *Note: the QSFP28 ports on the QFX5120-48Y can only be channelized into four channels, meaning that you cannot channelize these ports into 10x10 Gbps ports.*

On the QFX5120-48Y The “S” ports are grouped in quads (groups of four, ports [0-3], ports [4-7], etc,) and, as such, the speeds of these ports can only be configured in quads. The SFP28 port speeds cannot be configured individually. For example, to set port 6 to 25GbE or 1GbE, you must configure the quad of ports 4-7 at the same speed for the configuration to take.

Figures 1-3 show an example from the Port Checker tool to visually see how the speed configuration should happen.

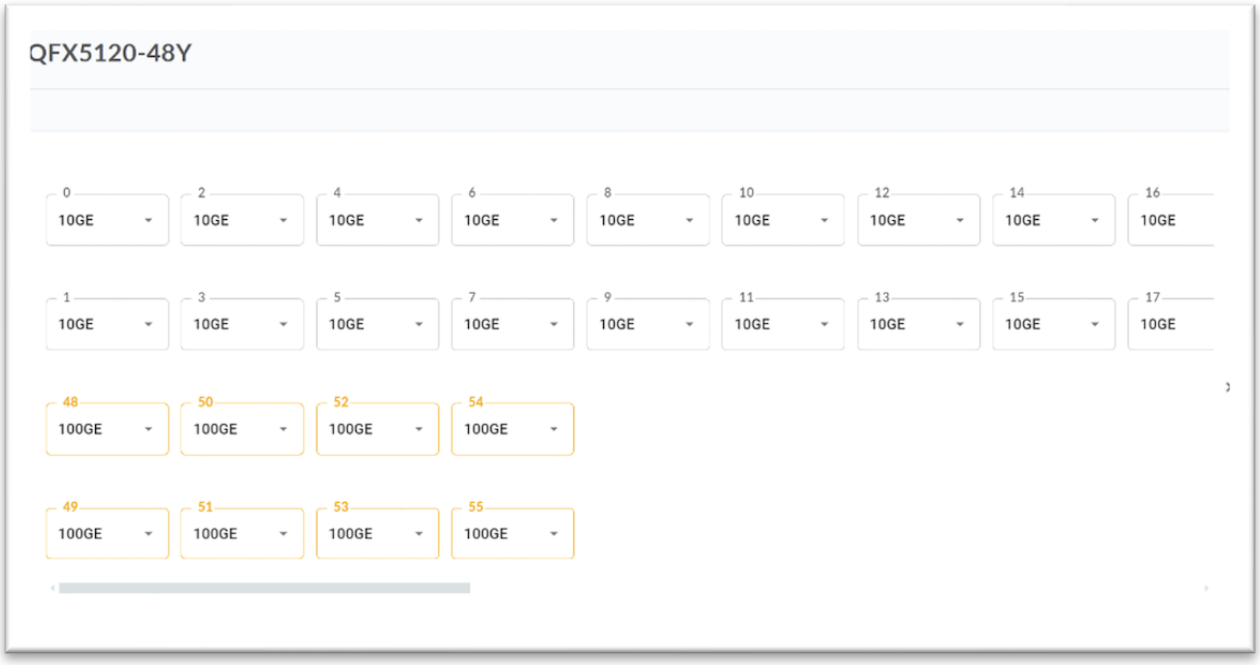


Figure 3: QFX5120-48Y at its default speed of 10G and 100G; a visual representation

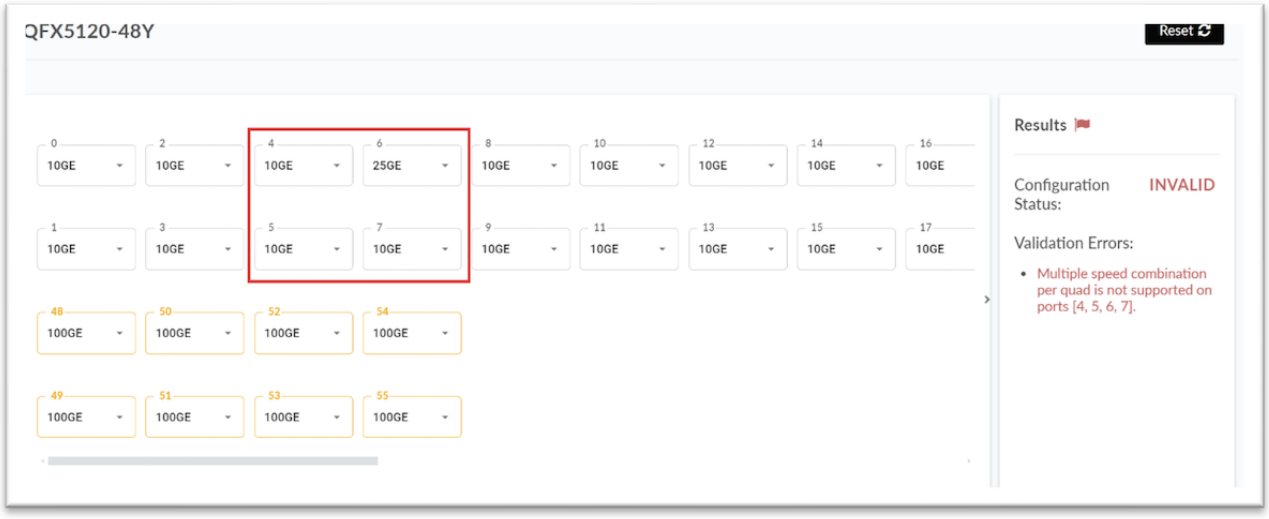


Figure 4: The QFX5120-48Y will not allow the configuration of individual speeds of the SFP28 interfaces; a visual representation.

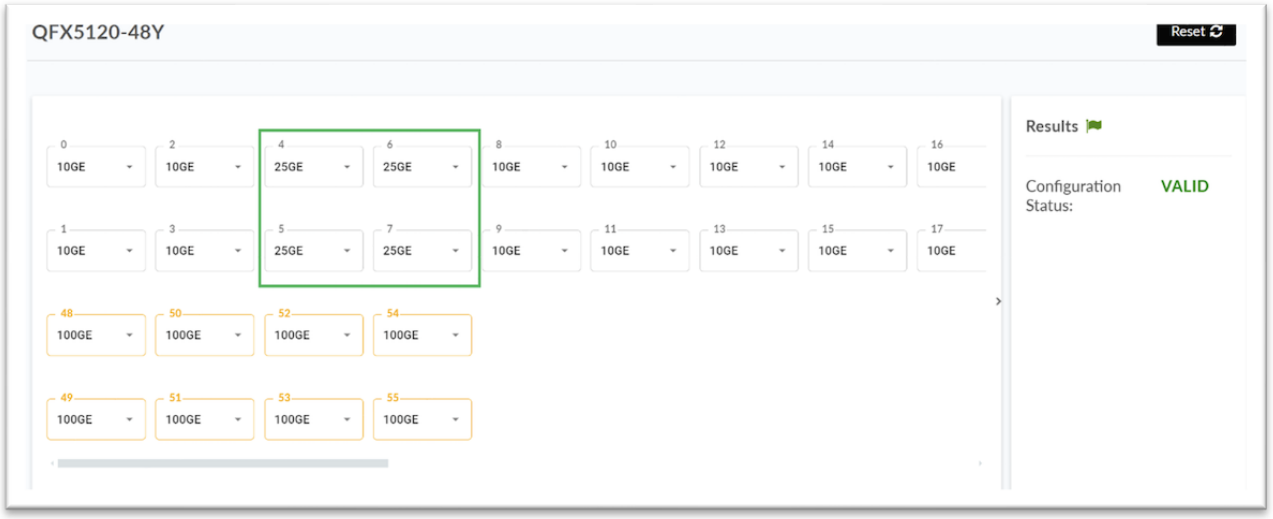


Figure 5: The QFX5120-48Y speeds of the SFP28 interfaces can only be configured in quads (ports 0-3, ports 4-7, etc)

Below is a configuration example to set port 6 on fpc 0 pic 0 on a QFX5120-48Y to 1 Gbps.

CLI format :

```
set chassis fpc <fpc_number> pic <pic_number> port-range <lower_port_number> <upper_port_number>
     channel-speed <speed>

#set chassis fpc 0 pic 0 port-range 4 7 channel-speed 1g
#commit
```

There is no commit check. If an individual SFP28 port speed is configured, the device will not give an error message. However, the speed will not change.

Below is a configuration example to change a configured quad of ports from one speed to another, you must delete the configuration first and then set the next configuration.

```
#delete chassis fpc 0 pic 0 port-range 4 7 channel-speed 1g
```

```
#set chassis fpc 0 pic 0 port-range 4 7 channel-speed 10g
#commit
```

On the QFX5120-48Y the speeds of the QSFP28 interfaces (ports 48-55) can be configured individually, unlike the SFP28 ports.

1.2.2 From QFX5100-48T to QFX5120-48T

The QFX5100-48T has 48 tri-speed copper downlink ports which are capable of 10 Gbps, 1 Gbps and 100 Mbps speeds using 8P8C (RJ-45)-based Category 5e or better ethernet cables, depending on the distance of the run. The QFX5100-48T also has 6 QSFP+ uplink ports capable of 40 Gbps speeds using QSFP+ modules

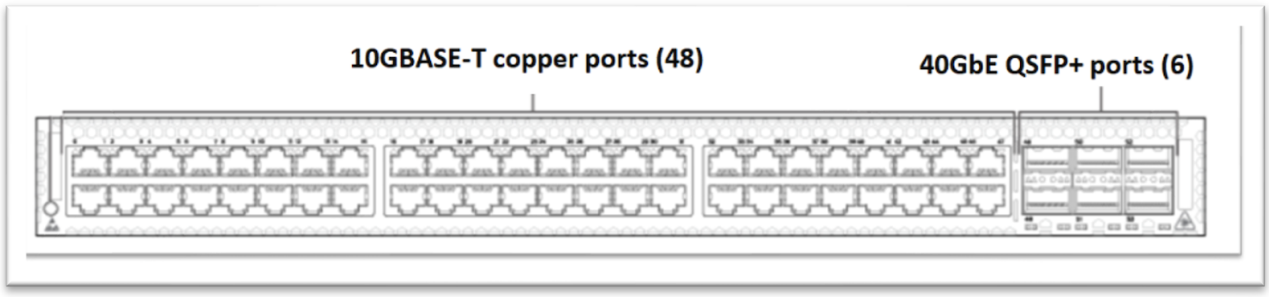


Figure 6: QFX5100-48T

The QFX5120-48T has 48 copper downlink ports which are capable of 10 Gbps and 1 Gbps speeds using 8P8C (RJ-45)-based Category 5e or better ethernet cables (though category 6 or better is strongly recommended), depending on the distance of the cable run. The QFX5120-48T also has 6 QSFP28 ports that support 100 Gbps and 40 Gbps speeds using QSFP28 modules for 100 Gbps and QSFP+ modules for 40 Gbps speeds.

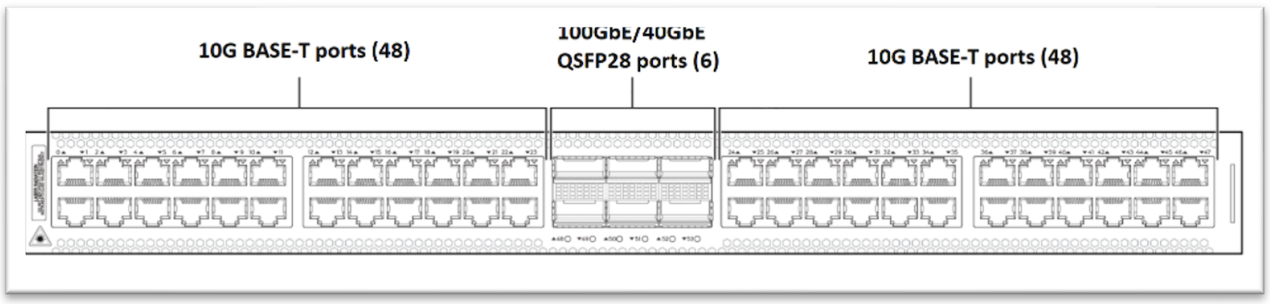


Figure 7: QFX5120-48T

Note: the QFX5120-48T does not support 100 Mbps speeds.

Port speeds on the QFX5120-48T are configured individually. For the 48 8P8C (RJ-45) interfaces, the default speed is 10 Gbps and can be set to 1 Gbps. These ports do not support channelization. The default speed of the QSFP28 interfaces (ports 48-53) is 100 Gbps. These ports do support channelization.

Channelization support of the QSFP28 interfaces on the QFX5120-48T is as follows:

- All QSFP28 ports support 2x50 Gbps
- Only QSFP ports 50 & 51 support 4x10 Gbps or 4x25 Gbps.

Below is a configuration example to set port 49 on fpc 0 pic 0 of the QFX5120-48T to 50 Gbps.

CLI format :

```
set chassis fpc <fpc_number> pic <pic_number> port-num <port_number> channel-speed <speed>

#set chassis fpc 0 pic 0 port-num 49 channel-speed 50g
#commit
```

The interface names of the 48 8P8C (RJ-45) ports start with “xe”, and the interface names of the 6 QSFP28 ports start with “et”.

Below is a configuration example to set port 6 on fpc 0 pic 0 of the QFX5120-48T to 1 Gbps.

CLI format :

```
set interface <interface name> speed <speed>

#set interface xe-0/0/6 speed 1G
#commit
```

1.3 QFX5100 to QFX5120: Software

1.3.1 Virtual Chassis

If you’re using Virtual Chassis (VC) functionality on your currently deployed QFX5100-series switches, be aware that the QFX5120-series switches do not support more than 2 members in a VC. If you have more than 2 members of a VC in your current QFX5100-series deployment you will need to engage your account team to discuss a redesign.

1.3.2 Channelization: Interface Naming Conventions

When a physical port supports multiple interfaces, a colon will be used in the interface naming conventions as a delimiter to distinguish the multiple interfaces on that physical port. In the interface naming convention, xe-x/y/z:channel:

- x represents the FPC slot number.
- y represents the PIC slot number.
- z represents the physical port number.
- channel represents the number of interfaces that are channelized.

When the 40 GbE interfaces (et-fpc/pic/port) are channelized as 10 GbE interfaces, the interface appears in the format of xe-fpc/pic/port:channel, in which “channel” is a value of 0 through 3.

A practical nomenclature example of this is channelizing the 40 GbE interface of FPC slot 0, PIC 0, port 6 into 4x10 GbE interfaces.

This would result in the interface name transforming from et-0/0/6 to xe-0/0/6:0, xe-0/0/6:1, xe-0/0/6:2, xe-0/0/6:3.

40GbE interface et-0/0/6 -> channelize -> 10Gbe interfaces xe-0/0/6:0-3

Table 1: Channelized and Non-Channelized Interface Naming Formats

Interfaces	Non-channelized InterfacesNaming Formats	Channelized InterfacesNaming Formats
10 GbE	Prefix is xe-.The interface name appears inxe-fpc/pic/port format.	Prefix is xe-.The interface name appears inxe-fpc/pic/port:channel format.
25, 40, 100, 200, 400GbE	Prefix is et-.The interface name appears inet-fpc/pic/port format.	Prefix is et-.The interface name appears inet-fpc/pic/port:channel format.

1.4 QFX5100 to QFX5120: Optics

Table 2: Optics supported for QFX5100-48S but not for QFX5120-48Y

Model Type	Model Number	Part Number	Description	Reasoning
40 GbE	JNP-QSFP-DAC-5MA	740-052306	QSFP+ 40GBase Direct Attach Copper Cable 5-meter, active	Lack of use case
40 GbE	QSFPP-40G-LX4	740-062254	QSFP+ 40GBase-LX4 40 GbE Optics, 100m(150m) with OM3(OM4) duplex MMF fiber	Please useJNP-QSFP-40G-LX4

1.5 Glossary

- 8P8C: 8 position, 8 contact. The proper name for the network plug is commonly referred to as “RJ-45”. In the context of computer networking, this refers exclusively to an unkeyed 8P8C connector supporting ANSI/TIA-568-wired cables.
- EOL: End of life
- FPC: Flexible PIC concentrator
- PIC: Physical interface card
- QSFP+: Quad small form-factor pluggable +
- QSFP28: Quad small form-factor 28
- RJ-45: Registered jack 45. This is a common term used for 8P8C connectors due to their similarity to the RJ45S standard. In the context of computer networking, this refers exclusively to an unkeyed 8P8C connector supporting ANSI/TIA-568-wired cables.

- SFP: Small form-factor pluggable
- SFP+: Small form-factor pluggable +
- SFP28: Small form-factor pluggable 28
- VC: Virtual chassis

1.6 Useful links

- QFX5100 documentation: https://www.juniper.net/documentation/product/us/en/qfx5100/junos-os/
- QFX 5120 documentation: https://www.juniper.net/us/en/products/switches/qfx-series/qfx5120-data-center-switches.html
- Pathfinder information for QFX5100 switches: https://apps.juniper.net/home/qfx5100/overview
- Pathfinder information for QFX5120 switches: https://apps.juniper.net/home/qfx5120-48y/overview

1.7 Acknowledgments

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